Tragedy of the Commons

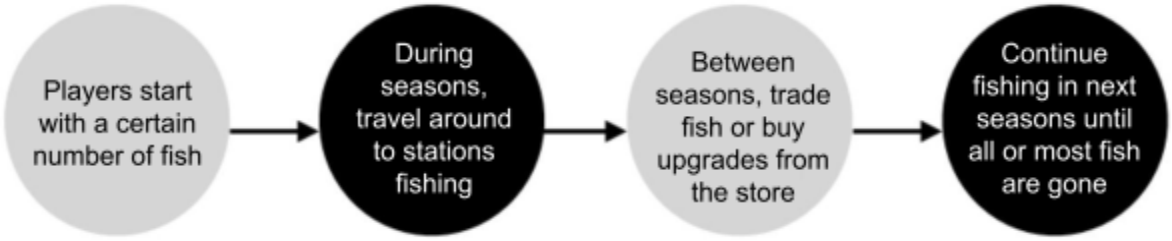
Facilitation Materials – Introductory Guide

Hurray, you've decided to run the Tragedy of the Commons Game (ToC)! Perhaps it will be an activity for the first day of camp, or a team-building program, or embedded as an integral part of your curriculum.

Whatever your objective may be, ToC is designed to be flexible and work across a wide range of settings, devices, and age groups. This **Introductory Guide** will give an overview of the game to help you prepare ToC for your specific context.

Overview

A single game of ToC typically flows as follows:



After an initial playthrough, players should discuss their experiences and generate hypotheses about why the fish disappeared, play the game again (repeat the above process) one or more times to test these hypotheses, and then reflect on what they've learned.

The game is based off the following mechanics/parameters:

Seasons	The "round" of fishing that the students are in. The seasons create a loop such that students go out to gather fish during the season and then come back to spend fish in between seasons.
Stations	Places where players can fish. Add or remove stations, name them, and choose the starting number of fish in each of them. They can be disabled before the game begins. In order to set up the classroom, you need to print the QR code for each station. For instructions, see Setting up the Stations under **Preparing to Run ToC**.

Trading	Toggle determines whether trading is available or not. If trading is activated, players can donate fish to other players during gameplay. This toggle can be changed midgame.
Local/remote gamemode	Toggle determines whether the game is played locally (face-to-face) or remote (online). This guide is focused on the local gamemode. For details about the difference between the two versions, see the Remote Documentation. After the game has been launched, this toggle can no longer be changed.
Statistics	Graphs that display and track the station populations over seasons, all players' numbers of fish, and the players with the top numbers of fish

The rest of this guide includes quick explanations of how to:

I. create and set up a game
II. run the game
III. facilitate a student-driven discussion
IV. end the game and facilitate student reflection

Sections I and II are covered in more detail in the Step-by-Step Guide.
Sections III and IV are covered in more detail in the Discussion Guide.

I. Preparing to Run ToC

<table>
<tr><td>Necessary Devices</td><td>For teachers:
Any device (e.g. phone, tablet, laptop, or desktop computer) that can connect to an Internet browser
A tablet or desktop is recommended</td><td>For students:
A charged, WiFi-connected mobile device with a camera and:
Android devices: Chrome browser
Apple devices [iOS 11 or later]: Safari browser
Fire tablets: Silk browser</td></tr>
<tr><td>Create Game</td><td colspan="2">The facilitator creates a game by going to the psims.games website, logging into the Host a Game homepage, and either creating a new game or selecting an existing one.</td></tr>
<tr><td>Setting up the Stations</td><td colspan="2">Create stations with your preferred starting number of fish. It is recommended to use the following combinations based on the number of players:
<table><tr><th># of players</th><th>Recommended starting setup</th></tr><tr><td><= 5</td><td>3 stations, ~50 fish each</td></tr><tr><td>6-10</td><td>3 stations, ~100 fish each</td></tr><tr><td>11-15</td><td>5 stations, ~150 fish each</td></tr><tr><td>16-20</td><td>5 stations, ~200 fish each</td></tr><tr><td>21-25</td><td>7 stations, ~200 fish each</td></tr><tr><td>26-30</td><td>7 stations, ~250 fish each</td></tr><tr><td>31-40</td><td>9 stations, ~250 fish each</td></tr><tr><td>41-50</td><td>10 stations, ~300 fish each</td></tr></table>
Select the QR code icon in the bottom right corner beside each station and print each one out. Display the printed station QR codes around the room, spreading them out.

Stations can be disabled before launching the game. Disabled stations can’t be scanned and won’t show up in graphs.</td></tr>
<tr><td>Being an effective facilitator</td><td colspan="2">As a facilitator, the two most important things to keep in mind are:
Give players as little information as possible for them to start playing.
Encourage players to observe, ask questions, and listen to each other.</td></tr>
</table>

Reminder: For a more detailed guide on how to set up the game, see the Step-by-Step Guide.

II. Running ToC

Start of the game	Have players go to **psims.games**, select **Join a Game**, and enter the class Game Code. When everyone is ready, select **Launch Game.** ● Once the game has been launched, it is not possible to change the game parameters, although new stations can be added. Then select **Continue Game.** ● This will start the game for the players. Players should start in the store by default at the beginning. Start the first season by switching players from the store to fishing mode.
During the game	During each season, players will travel between stations and scan the QR codes to fish in them, receiving a certain number of fish each time. Every time they fish in a station, they will have a five second scanning timeout and use up limited fuel in the season. To end the current season, move players back to the store. Between seasons, players have the opportunity to buy upgrades in the store and/or trade if it is enabled. After giving players sufficient time, press the **Next Season** button to move onto the next season. This will: ● double the population of fish at every station. ● deduct players' number of fish by the seasonal cost of 20 fish. ○ This deduction can cause players to go into debt with their number of fish, but the game is over for them if they have less than -20 fish.
End of the game	When all (or most) of the fish are gone, end the game by not advancing to the next season and asking players to take their seats.

Reminder: For a more detailed guide on how to run the game, see the Step-by-Step Guide.

III. Facilitating Discussion

After this initial game is over, the class should have a student-driven discussion focused on the concepts and questions that players want to explore in the next game.

The job of the facilitator is to initiate a student-drive discussion while pointing out concepts and questions that ar eimportant for players to explore in the next game. To do so, a facilitator should:

- Record notes on a chalk/dry-erase board
    - Try organizing notes as observations, questions, guesses, ideas, proposed experiments
- Names the target concepts and key terms as players describe them:

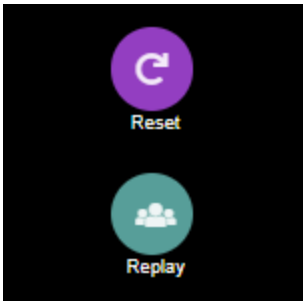
Target Concepts and Key Terms	Population growth rate Harvest rate Population size Initial population Maximum sustainable yield Quota Exponential Rate	During the discussion, players will describe many of these concepts without necessarily using the exact terms listed here. The job of the facilitator is to give names to these core learning goals as players "discover" them.

The facilitator *embraces all players' ideas* (even those that are off track) and *gently guides players to a consensus* about how the game will be played in the next game. A discussion should shift from observations about each **individual's experience**, through a process of analyzing players' **collective experience**, to a proposed experiment that will answer **players' questions.**

Reminder: For a more detailed explanation of how to facilitate discussion in ToC during the game, see the Discussion Guide.

IV. Wrapping Up

Once the discussion is over, the game should be played again one or more times to allow students to experiment. There are two buttons in the sidebar of the initialization dashboard that can quickly initiate a new game:



Reset: Removes all players and stations and reset all game parameters to their default values. Commonly used to start a new game with new players.

Replay: Keeps players and stations but revert player fish amounts and station fish populations to their initial amounts. Commonly used to quickly restart while keeping everything constant.

After the full game is over, the class should reflect on what they've experienced.

The purpose of the reflection is to help players learn that systems are made up of many parts, and problems are solved by many people, each with different perspectives. Additionally:

- Each player's participation in important – both in playing the game and in discovering how it works.
- The concepts this explores are important and connect to disciplines beyond ecology and population biology.

Reminder: For a more detailed explanation of how to facilitate reflection about ToC after the game, see the Discussion Guide.